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8 Not Enough Water

Drought Emergencies in the American Southwest

Melanie Gall

At the end of 2022, western states were on the brink of a water crisis. Lakes Mead and Powell, the largest storage reservoirs in the Colorado River system, had reached record low water levels, dropping to 25 percent of their storage capacity. The entire Lower Colorado River system was filled to only a third of its capacity.[1] Photographs of dry marinas, boats moored ever further away from the shorelines and previously flooded features, such as sunken boats, were in every newspaper. On the horizon loomed severe water cuts to Lower Colorado River Basin states that were previously unimaginable. The possibility of reaching dead pools in Lakes Mead and Powell, meaning water levels falling so low that water can no longer flow downstream across the dam and dams can no longer produce hydroelectric power, became a possibility. Eric Balken the director of the Glen Canyon Institute, went as far as saying that "we could be watching Lake Powell's mortality in real time."[2]

The Crisis Context

This chapter highlights the susceptibility of an essential natural resource – water – to climate change, population growth, and land use, and how water shortage and drought (a prolonged water deficit in the hydrological cycle) could disrupt American society and economy. For the purpose of this chapter, the focus will be on the ongoing megadrought in the West and the Colorado River Basin given that the Bureau of Reclamation issued the first-ever water shortage declaration (Tier 1) for the Lower Colorado River Basin on August 16, 2021.[3] This water shortage declaration went unnoticed by most and would likely not register among emergency managers as a crisis. But for some, especially farmers, it disrupted and nearly upended livelihoods (Figure 8.1).

There are many parallels between drought as a disaster and heatwaves. Drought, like extreme heat, falls outside of traditional emergency management processes. There has never been a presidential disaster declaration for a drought. While there are avenues for mitigation funding through the Federal Emergency Management Agency (FEMA), drought disaster response is generally managed by state agencies in charge of administering water resources. In the case of the Colorado River basins, state agencies such as Departments of Water Resources operate in conjunction with, or perhaps better under the purview of, the Bureau of Reclamation with predetermined drought contingency plans taking effect according to prior agreements among stakeholders. These drought contingency measures are comparable to preparedness measures such as hurricane evacuations where residents evacuate in a phased approach based on risk zones and evacuation times. Now imagine that these preparedness measures averted a crisis for nearly all residents with only a few, highly vulnerable groups, ultimately experiencing the physical impacts of the hazard. That is what

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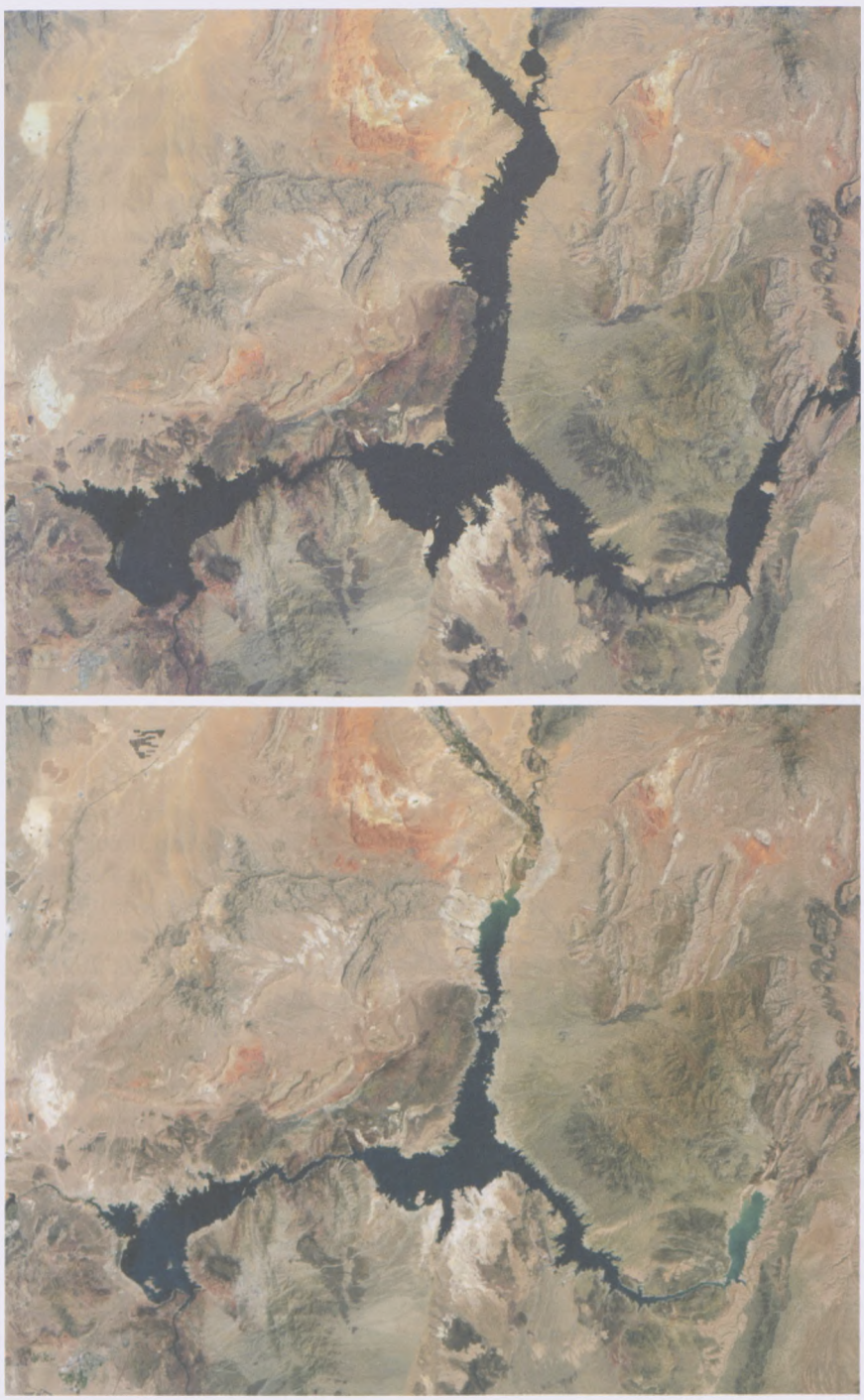


Figure 8.1 Satellite Imagery of Lake Mead. The top image was captured on July 6, 2000 while the bottom shows the reduction in lake levels on July 3, 2022.

Source: NASA, Carlowicz, Michael, and Lauren Dauphin. 2022. "Lake Mead Keeps Dropping." 2022. https://earthobservatory.nasa.gov/images/150111/lake-mead-keeps-dropping.

happened during the Tier 1 water shortage declaration. Not a disaster in the traditional emergency management sense, nevertheless catastrophic to those affected.

Water rights struggles among U.S. states and Native American tribes is part of the story and the slow onset of the 2022 water crisis, also illustrates the implementation of drought preparedness and response measures for different stakeholders. At present, the underlying challenges surrounding water dependency and water use by communities in arid areas, remain unresolved and point to future challenges as the compounding and cascading effects of dry hazards – heat, drought, and wildfire – unfold unabated and the need for integration into emergency management processes and equal access to resources remain. The near future will show if the drought crisis of 2022 was truly averted or if the water shortages woes of the West will expand beyond farmers and the agricultural sector. At the time of writing this chapter, the wet winters of 2022/23 and 2023/24 offered a short-lived reprieve and allowed the Colorado River system's storage capacity to recover somewhat and reach almost half-full capacity.[4] California's State Water Project (SWP) allocated 100 percent of water requests in 2023. This was the first time since 2006.[5] For a discussion of the various definitions of drought, types of droughts (meteorological, hydrological, agricultural, ecological, societal), and drought thresholds, the reader is referred elsewhere.[6]

More Dry than Wet Years

The U.S. Census Bureau defines the West as a combination of coastal states (Alaska, California, Hawaii, Oregon, and Washington) and mountainous states (Arizona, Colorado, Idaho, Montana, Nevada, New Mexico, Utah, and Wyoming).[7] The region's main watersheds drain to the east (the Missouri River, the Platte River), south (the Rio Grande), southwest (the Colorado River), and west (Columbia River, the Klamath River) along with the Yukon River in Alaska.[8] The Colorado River drainage basin encompasses the Colorado River and its major tributaries (the Green, the San Juan, and the Gila Rivers) (Figure 8.2). The point of separation of the Lower and Upper Colorado River Basin is Lees Ferry, Arizona, which is 15 miles downstream of Glen Canyon Dam and Lake Powell. The Upper Colorado River Basin produces approximately 90 percent of the annual stream flow with the vast majority of it stored in the snowpack of the Rocky Mountains.[9] According to the U.S. Bureau of Reclamation,[10] the average annual natural flow of the Colorado River is 14.7 million acre-feet (MAF) based on the period of record from 1906 to present (Table 8.1). This annual water yield has fallen to 12.4 MAF since the turn of the century. Both values are well below the annual allotments of the river's water to its users.

In 1922, the Colorado River Compact, an interstate agreement, established the division of Colorado River waters between states of the Upper Division (Colorado, New Mexico, Utah, and Wyoming) and states of the Lower Division (California, Nevada, and Arizona, the latter of which joined after 1922). Both divisions have equal rights to half the annual yield of water in the river, with an additional MAF allocated to the Lower Basin because Arizona argued it "deserved the use of the flow of the Gila River."[11] The ultimately established and apportioned water of 17.5 MAF (7.5 MAF to each basin and 1.5 MAF to Mexico) was determined during a period of high discharge with a limited understanding of the river's high flow variability and actual annual yield.[12] According to the Bureau of Reclamation, the basin-wide average long-term historical natural flow is steadily declining (Figure 8.3). The river's water yield has become "the victim of wishful thinking."[13] Today, water rights and allocations among the Colorado Compact members are governed by interlocking laws, treaties, regulations, guidelines, and court decisions such as the 1928 Boulder Canyon Project Act, the 1944 U.S.-Mexico Water Treaty, the 1948 Upper Colorado River Basin Compact, the



Figure 8.2 Map of the Colorado River Basin Boundaries

Source: U.S. Bureau of Reclamation, www.usbr.gov/lc/images/maps/CRBSmap.jpg

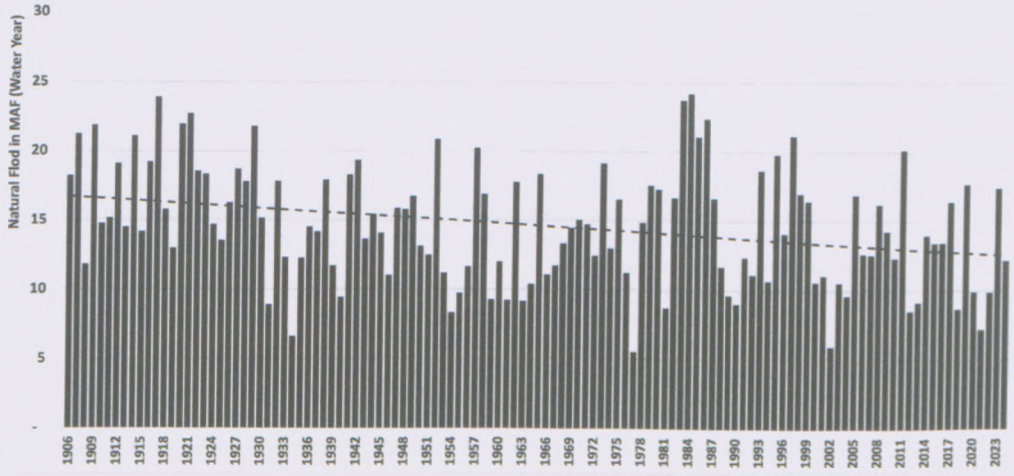


Figure 8.3 Average Annual Water Yield (Per Water Year) of the Colorado River. Measured at Lees Ferry, Arizona at the beginning of the Grand Canyon, the trend line shows a steady decline in water yield since 1906.

Source: Graphic produced by M. Gall. *Data Source:* U.S. Bureau of Reclamation. 2024. "Colorado River Basin Natural Flow and Salt Data," April 22, 2004, www.usbr.gov/lc/region/g4000/NaturalFlow/provisional.html.

1956 Colorado River Basin Project Act, the 1964 Consolidated Decree of the Supreme Court in Arizona v California, the 1968 Colorado River Storage Project Act, and the 2007 Colorado River Interim Guidelines for Lower Basin Shortages and Coordinated Operations for Lake Powell and Lake Mead.[14]

Periods of low precipitation are common occurrences in the arid West (see Chapter 9). Present-day conditions, however, are approaching a tipping point. Since the turn of the century, the American West has been in a megadrought (Figure 8.4) – with the Southwest and the Colorado River Basin (about 246,000 square miles) particularly hard hit. Most recently, about half of the western

Table 8.1 Hydrological and Legal Terms and Definitions

Natural Flow	The natural flow of a river represents its unaltered, natural character with all its dynamics. Every river has a unique flow regime characterized by the magnitude, frequency, duration, timing, and rate of change of water flowing downstream. Magnitude represents the volume of water passing a fixed location along the river per unit time (see MAF below). Floods and droughts are part of a river's natural flow representing maximum and minimum magnitudes of flow.
	Frequency captures how often a flow above a specific magnitude recurs, such as a 100-year flood which has a probability of 0.01 of happening every year. Duration measures the time for which a specific flow volume persists (e.g., number of days during which 10-year floods are present). The timing of flows refers to the predictability or seasonality during which specific flows are prevalent. The rate of change expresses how quickly a river's flow can change. Streams with a rapid rate of change are called "flashy." Dams, diversion channels, and similar infrastructure alter a river's natural flow regime.
	Source: Poff, N., LeRoy, J., David Allan, Mark B. Bain, James R. Karr, Karen L. Prestegaard, Brian D. Richter, Richard E. Sparks, and Julie C. Stromberg. 1997. "The Natural Flow Regime." *BioScience* 47 (11) (1997): 769–784. https://doi.org/10.2307/1313099.

(*Continued*)

Table 8.1 Continued

Million acre-feet (MAF)	The terms flow, discharge, streamflow, or flow rate are used interchangeably and generally expressed in terms of volume per time. The most common units are cubic feet per second (cfs) or acre-feet per year. The unit most frequently used in this chapter is the annual flow of the Colorado River measured in Million acre-feet (MAF).
Water Yield	Water yield represents the quantity of water, usually measured as volume per time (e.g., acre-feet per year) that can be harvested from a unit area of watershed. Water yield can refer to groundwater, rivers, lakes, etc. A perennial water yield is the maximum amount of water that can be sustainably withdrawn from an aquifer without overdrafting/lowering the groundwater table permanently. *Source:* ASCE. *Hydrology Handbook: Second Edition.* 2nd ed. New York, NY: American Society of Civil Engineers (ASCE), (1996). https://sp360.asce.org/personifyebusiness/Merchandise/Product-Details/productId/231964613.
Water Rights	There is competition for water (surface water and groundwater). To ensure availability and access to water at specific places and satisfy demand by different stakeholders (e.g., municipalities, agriculture, industry, recreation, etc.), access to water is legally protected. A water right is the legal right to take possession of water meaning to use or divert water at a specific location from a natural waterway.
Water Law	Water law governs the use of water and differentiates between public and private rights, how these rights compare or are prioritized along with aspects of water quality and water pollution. Each state tends to have its own regulatory system for monitoring and controlling water use (surface and groundwater) including the construction of diversion systems, wells, etc. States tend to govern and regulate water allocations. Federal law governs water quality. The law governing groundwater rights varies among states. Rainwater harvesting regulations vary among states. *Source:* National Agricultural Law Center. "Water Law Overview." 2024. https://nationalaglawcenter.org/overview/water-law/.
Water Allocations	The maximum amount of water that is allocated to a stakeholder (also called allotment or apportionment) is quantified as part of a stakeholder's water right. There are three different allocation systems in the United States with variations among U.S. states: *riparian doctrine* (most common in the eastern U.S.), "first-in-time, first-in-right" (most common in the western U.S. and discussed in this chapter, also called *prior appropriation doctrine*), and hybrid systems consisting of both riparian doctrine and "first-in-time, first-in-right." Riparian water rights ensue with the possession of land adjacent to a body of water. Riparian water rights allow for reasonable use of water (e.g., drinking, irrigation, watering livestock, industrial use, etc.) without infringing on the water rights of a landowner downstream. Most states with riparian rights use a permitting system to regulate water rights. Within a prior appropriation system, the member who was first in, has the highest priority and may use up all their allocated water amount without any requirement to use less water in time of shortage or leaving any water to junior members. The Colorado River Compact is such a system. Groundwater may be allocated based on the Absolute Dominion Rule, Correlative Rights Doctrine (e.g., California), Prior Appropriation Rule (most western states), Reasonable Use Rule (e.g., Arizona), or the Restatement (Second) of Torts Rule. *Source:* National Agricultural Law Center. "Water Law Overview." 2024. https://nationalaglawcenter.org/overview/water-law/.

region experienced either exceptional (20 percent) or extreme (30 percent) drought conditions for 37 weeks starting at the end of 2020 and for most of 2021 followed by wet winters in 2023 and 2024 allowing reservoirs and groundwater levels to rebound slightly.

These persistent and anonymously dry and warm conditions in the Southwest have only been exceeded in severity and duration by the megadrought of the late 1500s. In combination with anthropogenic warming though, it appears this megadrought is on a "trajectory comparable to the worst southwestern North America megadroughts since 800 CE."[15]

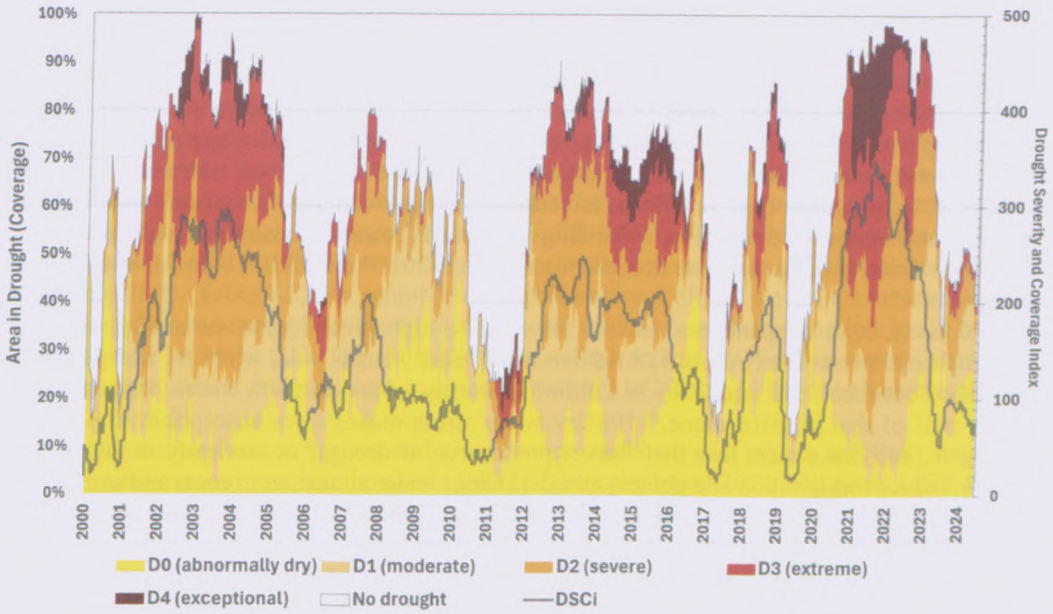


Figure 8.4 Weekly Timeline of Areas in the American West Under Dry or Drought Conditions. The secondary axis, Drought Severity Coverage Index, translates drought severity and area covered by that drought category into a single index that allows for drought comparisons over time.

Source: Graphic produced by M. Gall. *Data Source:* Drought Monitor. 2024. "Data Tables." June 25, 2024, https://droughtmonitor.unl.edu/DmData/DataTables.aspx?dregion,8.

Compounding Dry Hazards

The megadrought is not "just" a drought. It is a multi-hazard phenomenon of cascading and compounding effects that exacerbate dry conditions and their impacts. The anthropogenic amplification, i.e., warming of 0.2°C ± 0.01°C per decade (daily maximum temperature) and annual precipitation reduction of -2.3 ± 1.5 mm per decade in the West[16] – along with the coupled consequences of dry hazards – drought, heat, wildfires, and one may add smoke pollution – are not just making the drought worse but are causing significant cascading natural and societal consequences previously unseen in their extent and magnitude.

First, extreme day- and night-time temperatures and prolonged heatwaves in the West are creating life-threatening conditions (see Chapter 7). Above-normal temperatures increase evaporation, winter evapotranspiration, and snow sublimation in the mountainous areas thereby reducing runoff into the Upper Colorado River Basin, which produces most of the water in the Colorado River.[17]

Second, although the megadrought of the West is characterized by deficits in precipitation, there are short, interspersed periods of extreme rain- or snowfall as seen in Chapter 9 and in the winters of 2023 or 2024. High precipitation variability[18] combined with less snowfall, shallower snowpacks, and less precipitation falling as snow make water supply predictions increasingly difficult.[19] Climate change is causing an earlier onset of snowmelt of shallower snowpacks along with slower snowmelt and lower meltwater volume due to infiltration (movement of water from the soil surface into the ground).[20] Musselman et al. found that "[w]idespread melt increases across western North America, even though lower change is observed in commonly used snow metrics, indicate that this critical water resource is in steeper decline than is inferred from (snow water equivalent) trends

alone."[21] Combined, these climate trends make drought and water management along with drought prediction significantly more challenging.

Third, dry conditions accelerate heatwaves.[22] Hot and dry conditions also pave the way for water-stressed forests and wildfires as the lack of precipitation and moisture accelerate soil and fuel drying (see Chapter 6). The higher the temperature, the shorter the drought period needed to stress vegetation. In a study, analyzing wildfire trends (timing of occurrence, frequency, size) from 1984 to 2011, Dennison et al. found that increasing trends in wildfires "were most significant for southern and mountain ecoregions, coinciding with trends toward increased drought severity."[23] Other research shows that extreme fire seasons such as in 2011[24] and 2020[25] coincide with drought conditions.

And lastly, high temperatures, smaller snowpacks, high precipitation variability, or earlier snowmelt can stress trees. About 128 million trees, mostly sugar pine, white fir, and ponderosa pine, died between 2012 and 2016 in California, mostly in the southern Sierra Nevada Range, nearly half of that in 2016 alone.[26] Hot/dry water stress makes trees susceptible to pests and pathogens, can make them lose their leaves prematurely if drought occurs early in the growing season, reduce transpiration and photosynthesis, change water allocation to roots and storage, and overall reduces nutrient uptake altering nutrient flow to the tree and soil.[27] Thus, droughts reduce carbon uptake and storage in trees.

Blurring the Line between Human-Made and Natural

While the compounding adverse effects of hot and dry conditions are well-established, the concentration, expansion, and emergence of (new) cascading environmental, economic, and societal adverse impacts under anthropogenic climate conditions are cause for concern and a sign of ineffective or maladaptive drought management practices. Among the dry hazards, droughts drive the magnitude and occurrence of cascading events.[28] To make matters worse, the anthropogenic amplification of the compounding dry hazards is not only making droughts more severe but is causing significant cascading natural and societal consequences previously unseen in their extent and magnitude.[29]

Consequently, the water shortage crisis in the Lower Colorado River Basin has moved from a hidden risk to outright threatening the livelihoods of those living in the Southwest. Water is life. But where do you draw water from when there is scarce rainfall and not enough surface water?

The Slow Onset of the 2022 Water Crisis

Today, the over-allocated Colorado River Basin supplies water to 40 million people and supports 5.5 million acres of farmland.[30] This was only achievable through massive investments into water delivery and storage systems. The construction of massive dams and reservoirs since the establishment of the Colorado River Compact in 1922 meant that the availability of water supercharged population, economic, and agricultural growth. The Bureau of Reclamation performs a balancing act of 1) maintaining adequate lake levels to generate hydropower through the Glen Canyon Dam at Lake Powell, the Hoover Dam at Lake Mead, and others, and 2) providing sufficient water flow down the Colorado River to meet water allocations to recipients in the Lower Colorado River Basin pursuant to the Colorado River Compact. The water storage projects in the Colorado River Basin can retain up to 60 MAF. Lake Powell in the Upper Basin can capture up to 25.1 MAF[31] and Lake Mead in the Lower Basin can store up to 28.9 MAF.[32]

The 2022 water crisis came with a long warning. Throughout the megadrought, the apportionments of water above the Colorado River's natural flow were made possible only by the water supply retained in Lake Mead and Lake Powell. These reservoirs insulated water users in the Lower Colorado River Basin from the river's declining and variable flows but it resulted in steadily decreasing water levels in both reservoirs throughout the 2000s. To curb water use, the seven Colorado River Basin states signed an "Agreement Concerning the Colorado River Drought Contingency Management and Operations" in 2019.[33] And pursuant to dam and reservoir operating guidelines developed in 2007 (72 FR 62272, FES-7–37), the Bureau of Reclamation implemented pre-determined, involuntary water delivery cuts to Arizona, Nevada, and Mexico in 2021 (Table 8.2). As senior water rights holder, California was not subject to involuntary cuts but encouraged residents to voluntarily reduce water consumption by 15 percent (Executive Order N-10–21). The state also required the activation of urban water contingency plans and agricultural drought plans to conserve water.[34] In 2022, additional voluntary cuts took effect (Table 8.2).

Still the unabated use of surface and groundwater combined with supply shortfalls setting resulted in a new record low lake level of 1,040.71 ft in Lake Mead 2022 (full pool: 1,229 ft).[35] In 2022, the three Lower Basin states agreed to conserve 3 MAF from 2023 through 2025 out of which 2.3 MAF in reductions were compensated with drought relief funds ($4 billion) from the federal Inflation Reduction Act to help farmers, communities, and Native American tribes to pay for these voluntary cuts.[36] With much of the water retained in the snowpack during the wet winter of 2022/23, the record low lake level (3,529.92 ft) of Lake Powell occurred on April 13, 2023 (full pool is 3,700 ft).[37]

Impacts of the 2022 Water Crisis

Nearly 60 percent of Colorado River water goes toward irrigated agriculture, 13 percent to municipal, commercial and industrial use, and 16 percent to riparian vegetation and wetlands. The rest is

Table 8.2 Drought Response Measures to Lower Basin States as Outlined in the 2019 Drought Contingency Plan and the 2007 Operating Guidelines*

Projected January 1 Lake Mead Elevation (in ft)	*2007 Interim Guidelines Shortages (involuntary) (in MAF)*		*2019 Drought Contingency Plan (voluntary) (in MAF)*		
	AZ (annual allocation: 2.8 MAF)	*NV (annual allocation: 0.3 MAF)*	*AZ*	*NV*	*CA (annual allocation: 4.4 MAF)*
1075.01 to 1090.0	0	0	0.192	0.08	0
1050.0 to 1075.0 (Tier 1)	*0.32*	*0.13*	*0.512*	*0.21*	*0*
1045.01 to 1049.99 (Tier 2a)	*0.4*	*0.17*	*0.592*	*0.25*	*0*
1040.01 to 1045.0	0.4	0.17	0.64	0.27	0.2
1035.01 to 1040.0	0.4	0.17	0.64	0.27	0.25
1030.01 to 1035.0	0.4	0.17	0.64	0.27	0.3
1025.0 to 1030.0	0.4	0.17	0.64	0.27	0.35
< 1024.99	0.48	0.2	0.72	0.3	0.35

* Water cuts implemented in 2022 (Tier 1) and 2023 (Tier 2a) are highlighted in italic type. In 2024, the system went back into Tier 1 shortage.

Source: U.S. Bureau of Reclamation, "Agreement concerning Colorado River Drought Contingency Management and Operations," May 20, 2019, www.usbr.gov/lc/region/g4000/dcpdocs/Companion-Agreement-Final.pdf

lost to reservoir evaporation.[38] In 2022 and 2023, this meant that farmers in Arizona bore the brunt of the water cuts (a 21 percent reduction) implemented by the Central Arizona Project (CAP) system, a 336-mile long storage (Lake Pleasant) and delivery system that conveys the Colorado water to Maricopa, Pinal, and Pima counties.[39]

Arizona's irrigated farming produces large shares of the nation's winter lettuce, alfalfa, durum wheat, pecans, cantaloups, chili peppers, dates, honeydew melons, spring onions, and pistachios along with cotton for export.[40] In Pinal County, AZ, where agriculture is a $2.3 billion business per year, agricultural water cuts were the most severe. There, irrigated farming had to cope with a 87 percent reduction of water allotments in 2022, and a 95 percent reduction in 2023. With many dairy farmers having relocated to Pinal County due to development pressures in Maricopa County, water cuts resulted in less feed, fewer farming equipment sales, and the loss of thousands of agricultural jobs.[41] Farmers resorted to fallowing land to curb their water needs, plant less water-intense crops or crops with a shorter growing cycle, lease their land (e.g., solar farms) or pump more groundwater – the latter of which has been the dominant drought response measure.[42] Without substantial investments into (ground)water distribution, more efficient water use and irrigation systems, and more adaptive farming practice, irrigated agriculture as a whole is at risk in Arizona as it is already heavily dependent on government subsidies, crop insurance, and other forms of disaster relief.[43]

But it is not only farmers who are impacted by the current water crisis. For example, the City of Scottsdale, Arizona terminated its decade-old water contract with the community of Rio Verde Foothills, a newer suburb with $500,000 homes outside of the city's limits. About 2,000 residents were essentially disconnected overnight from trucked-in water from the City of Scottsdale. A costly struggle ensued for these homeowners to secure and pay for a scarce resource: trucked-in Colorado River water from elsewhere. As a consequence, homeowners had to cut their water use (fewer showers, more usage of paper plates, etc.), incurred water bills higher than their mortgage payments, or were forced to sell their homes, the latter the ultimate adaptation to the risk.[44]

A Second Water Crisis in the Making

The Southwest's water emergency is also a groundwater crisis, in part. Among the most common drought response practices is the use of groundwater but opportunities for groundwater renewal have been scarce over the past decades (e.g., 2017, 2019, 2023). Reduced precipitation caused not only lake levels and river discharge to fall across the Colorado River Basin[45] but also for groundwater levels to fall precipitously either naturally due to lack of infiltration or anthropogenically due to groundwater pumping for public water and farming purposes.

Reductions to Colorado River allocations mean cuts to farming, which consequently leads many farmers to pump groundwater. In California, where, for example, the drought years from 2012 through 2016 led to largely unregulated groundwater extraction. Falling groundwater levels triggered a host of unwanted consequences such as dry wells, subsidence including subsidence of water canals such as the Delta-Mendota Canal and Friant-Kern Canal in California, saltwater intrusion in coastal aquifers, concentration of groundwater contaminants as well as the flow of pollutants to adjacent areas, or siphoning off water from surface streams and groundwater.[46] The extent to which groundwater tables have declined in California and domestic and agricultural wells have run dry is largely anecdotal and dependent on voluntary reporting. Estimates suggest that in California's Central Valley, as many as one in five wells run dry[47] despite California's passage of the Sustainable Groundwater Management Act in 2014 (AB 1739, SB 1168, and SB 1319) and the creation of Groundwater Sustainability Agencies in California to avoid over-drafting within 20 years.

Groundwater is less stringently regulated in Arizona than in other Colorado River Basin states, some might even say lax. About 80 percent of the state does not regulate groundwater use. It is therefore unsurprising during a megadrought that groundwater levels have fallen in the nearly 60 percent of its 51 groundwater basins.[48] The 1980 Arizona Groundwater codes regulates the management of groundwater within six Active Management Areas (mostly densely populated communities) – all of which, however, have drawn more groundwater than replenished it (with the exception of 1993) hereby exceeding the stated goal of "safe yield."[49] The 1986 Underground Water Storage and Recovery Act and the 1994 Underground Water Storage and Replenishment Act along with the Arizona Water Banking Authority (AWBA) aim to prevent groundwater depletion and bank water to counteract annual variability. As stated elsewhere, "Although Arizona laws closely control the recovery of recharged water, the abstraction of other water in the aquifer is governed by other laws that do not fully restrict pumping. Thus, groundwater levels may continue to drop in locations where recharge has occurred."[50] In fact, they are, as anecdotal newspaper reports reveal, often pitting residential water use versus agricultural/rural use. As a lobbying group puts it "Alfalfa farms, along with the proliferation of mega-dairies, are sucking the Colorado dry – and the West's lifeblood alongside."[51] If Arizona wants to avoid draining its aquifers, massive changes in practices are on the horizon for irrigated agriculture and dairy farming.[52]

Native Water Rights

Tribal nations have neither had a seat at the table during the 1922 Colorado River Compact negotiations nor were their water rights established. Tribal water rights are presumably senior (i.e., oldest and highest priority) to other users but many of the 30 federally recognized tribes in the Colorado River Basin have ongoing or unresolved litigation regarding tribal water rights, often dating back decades.[53] Twenty-two out of the 30 tribes have recognized water rights to use up to 3.2 MAF, which together with any pending water rights would reduce the apportionments to the U.S. states.[54]

Some tribes had portions of their water rights already adjudicated while other portions remain in litigation. The most recent case of *Arizona et al. v. Navajo Nation et al.* decided by the Supreme Court on June 22, 2023, ruled against the Navajo Nation siding with the United States that the U.S. is not required to quantify water allocations to the Navajo Nation given that the 1868 treaty did not enunciate water demands. Instead, the Court suggested that Congress should act to enshrine water rights, access, and infrastructure for Native nations.[55]

Receiving allocations and utilizing the apportioned water rights, requires not only legal standing but also delivery and storage infrastructure – expensive infrastructure built over decades. The Central Arizona Project cost $4 billion dollars, and the California State Water Project (SWP) includes more than 20 dams and reservoirs (e.g., Lake Oroville, Pyramid Lake) and relies on a massive system of canals, aqueducts, and pumping stations. Many tribes lack funding for such infrastructure or have outdated and/or inefficient delivery systems.

Adaptation to Hotter and Drier Futures

As stated elsewhere water management must equally include groundwater management and that technological, scientific, political, and governmental innovations are urgently needed to protect and sustain groundwater and surface water for future generations.[56] This is particularly true for the current water crisis in the West – an all-hands-on-deck approach is needed and all options should be on the table.

The water crisis of 2022 has certainly elevated the issue of water management in Colorado River Basin states and some movement in the right direction is occurring. However, overcoming water supply challenges such as those in California, where "groundwater use accounts for 41 percent of the State's total water supply on an average annual basis but as much as 58 percent in a critically dry year, and approximately 85 percent of public water systems rely on groundwater as their primary supply,"[57] is a daunting task and will require massive investment into small and big projects, as well as difficult choices and a change in behavior.

Among the easier to achieve actions, particularly when incentivized, are water conservation and efficiency measures such as smart water meters (including metering to separate indoor from outdoor use), xeriscaping (desert landscaping), limiting non-agricultural irrigation (in effect in NV as of June 26, 2024), installation of drip irrigation for rural and agricultural use, water banking, and more. However, the effectiveness of the latter to manage water remains murky.[58] In the City of Phoenix, such efforts in addition to others, have resulted in a reduction of water consumption rates of about 30 percent over the past twenty years.[59] No mandatory water use reduction policies have ever been put in place in Phoenix. Below the surface though, water access is already creating discrepancies. A few (less than 1 percent) of Phoenix' residents have retained an outdated and water-guzzling practice: cheap, untreated water for flood irrigation, where yards are flooded with 2 to 3 inches of water to soak the soil deeply.[60] In Phoenix, this separation between water-rich and water-poor neighborhoods correlates directly with hotter and cooler neighborhoods double-burdening the more vulnerable and less well-off (see Chapter 7).

At the governmental level, integrated and collaborative regional water management as well as enforcement will become critical for a sustainable future. In 2023, for example, the City of Tucson, AZ, entered into a water-sharing agreement with sister cities in central Arizona to store portions of their allocated Colorado water in Tucson's Southern Avra Valley Storage and Recovery Project facility.[61] This shows how municipalities are seeking and developing infrastructure-based water banking/storage solutions to insulate their residents against water shortage. Another example of cooperation, albeit not a true seat at the table, is the memorandum of understanding between the Upper Colorado River Commission and the Jicarilla Apache Nation, Navajo Nation, Ute Mountain Ute Indian Tribe, Southern Ute Indian Tribe, Ute Indian Tribe and the Shivwits Band of Paiutes that was proposed in March 2024. The MOU calls for bi-monthly meeting with the intent of providing a cooperative structure and participation as all Colorado River stakeholders work toward a post-2026 operational plan for the Colorado River.[62]

More difficult, in terms of investment needed and/or public perception (i.e., financially and politically challenging), are actions such as restricting and limiting new groundwater wells, monitoring and limiting groundwater pumping, the recycling and purifying of wastewater (so-called reclaimed water), desalination, and expanding the water delivery and storage infrastructure to transport non-Colorado water or to capture water from intense rainfall or flood events.

In its most recent water plan, the State of California boldly admits that "California's existing flood, water, and wastewater management and treatment systems will be unable to deliver adequate, much less the same level of, quality and service as in the past" and that "California's ability to adapt is not keeping pace with climate change impacts."[63] To strive toward watershed resilience, the state is committing, among many other actions, to new infrastructure investments. The State of California is proposing, for instance, to expand its water supply through new projects such as the Delta Conveyance Project, the Sites Reservoir Project and Flood-Managed Aquifer Recharge Programs to capitalize on the co-occurrence of floods and droughts in the state.

Perhaps the strongest evidence regarding the willingness to adapt lies in the past two decades. The current water emergency did not occur overnight. In Arizona, a Drought Emergency Declaration (PCA 99006) has been in place since June 1999 and a Drought Declaration (Executive Order

2007–10) since May 2007. In Arizona, farmers have shouldered water cuts while residential use remained unaffected though change may be in the offing. Arizona's groundwater woes resulted in a recent moratorium on the development of new homes relying on groundwater.[64] Additionally, Governor Hobbs canceled land leases (and access to groundwater pumping) with a foreign-owned Alfalfa.[65] On the other hand, the Governor signed an emergency law (SB 1432) that forced the City of Scottsdale to reinstate water supply to Rio Verde through a new standpipe for three years. The development loophole that blindsided residents in the first place, however, remains in place and "wildcat" subdivisions are still legal. Water problems are plentiful in Arizona and elsewhere but solutions remain elusive.

Not everyone is equally affected by adaptation strategies or the lack thereof. Frontline communities, i.e., those experiencing lack of water or water of poor quality first (and possibly worst) are most exposed to water shortages and affordability issues. To buffer some of these inequitable burdens, the federal government is investing $12.9 million over 5 years for water conservation, water infrastructure projects, water purification and reuse, water storage and conveyance, desalination and dam safety through the Bipartisan Infrastructure Law and the Inflation Reduction Act.[66]

What the Future Holds

The entire American West remains in a megadrought. Nowhere else do the drought conditions and water demands as critically as in the Southwest. At the time of writing this chapter, California has lifted its drought emergency, Arizona's drought declaration remains, and Nevada has newly re-issued a water shortage emergency asking residents to reduce consumption by 20 percent.[67] The Colorado River operations remain in the first-ever Tier 1 shortage declaration and water recipients are jockeying to inform the development of future operations of the Colorado River as several agreements and the 2019 Drought Contingency Plans are set to expire at the end of 2026.

The post-2026 guidelines will determine the operations of Lakes Mead and Powell for years to come. The amount of water that will be delivered under various reservoir conditions to the lower basin states is at stake. What has thus far been labeled "drought-related water cuts" could become the new normal water allocations post-2026. For Arizona, whose water allocation has lower priority than California and Nevada, and where 36 percent of its water budget comes from the Colorado River, this could mean significant, permanent cuts under the current megadrought conditions.[68] California, on the other hand, has more financial resources and more water sources including the potential to receive and harvest occasional vast sums of precipitation. Given the massive snow and rainfall amounts in 2022/2023 and 2023/24, California is eying structural investments into conveyance and storage projects.

The lower basin states have shown that they can reduce their reliance on Colorado River water, but it is unclear to which degree permanent cuts will adversely affect agriculture and groundwater levels in these states. If past is prologue, (ground)water solutions will remain elusive until it is too late. The time to institute adaptation practices (i.e., water-saving infrastructure, conservation practices, regulation, behavior changes) is when there is still water. Experts such as Jason Robinson are pessimistic and imagine that "any major policy changes are probably still a long way off" while conceding that "[e]veryone recognizes that the patient is bleeding out on the table."[69]

It is both questionable and concerning that the past wet winters may not terminate the megadrought that the West has been engulfed in. As Littell puts it: "The additive influence of increasing temperature associated with climate change exacerbates meteorological droughts of historical magnitude."[70] Others are more candid and estimate that annual flow to the lower basin may decline by 33 percent, thus more urgent measures than involuntary and temporary, voluntary water cuts are needed.[71]

Despite the historic and projected declines in water yield from the Colorado River, the apportionments established by the Colorado River Compact remain unchanged. Instead, the engineering and accounting of every drop in the Colorado River Basin continues to this day as exemplified by the Bureau of Reclamation's ongoing Colorado River Storage Project.

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Postscript

Drought

The desert Southwest has always been somewhat inhospitable to settlement with water being the scarcest resource. As this chapter highlights, droughts are commonplace in this region, but with the addition of climate change, and bustling population growth, the socioeconomic impacts of droughts are reaching tipping points in many places. How sustainable are Las Vegas and Phoenix in the future?

As the megadrought in the Colorado Basin took hold in the last twenty years, the steady decline in water yield from the Colorado River occurred as the chapter points out. At the same time, Phoenix experienced rapid growth, increasing its population by 25% since 2000 based on U.S. Census data. At the same time, Maricopa County (including Phoenix and its suburbs) grew by 49%. But Phoenix was not the only desert city to experience exponential growth. Las Vegas started the century with a little more than 475,000 people, but by 2023 it had grown by 38% to more than 660,000. Just like the Phoenix-Maricopa County relationship, the suburbs of Las Vegas also swelled by 70% to 2.3 million people since 2000 increasing its own demand for water and power from the Colorado River.

The physical expansion of these two metropolitan areas has fueled the need for additional demands for residential water but also power. One of the cascading effects of the lower levels of Lake Mead, was the real fear of reduced hydroelectric generation from the Hoover Dam – the primary energy source to power the bright lights of Las Vegas. Hoover Dam provides electricity for more than 1.3 million people in Nevada, Arizona, and California and as the water levels dropped there were fears of brownouts to reduce electrical demand. Yet the relationship between population growth and water demand is not linear and while population increases may increase by 45% in the Phoenix region, water demand only increases by 14% in the same area due to water efficiencies, conservation, and landscaping choices.

As the chapter clearly illustrates, cities are not the only user of water and power from the Colorado River. The agricultural sector uses 80% of the water allocation to irrigate crops. When droughts occur, not only do people suffer directly, but it reduces food resources, potentially prompting food scarcity for people and animals. The Yuma, Arizona area grows much of the nation's winter supply of leafy greens so a reduction in surface water allotment means fewer salad ingredients are produced, which raises prices in grocery stores. The experience with the megadrought and the declaration of a water shortage prompted changes in surface water allocation and a re-focusing on drought emergency planning as the chapter points out. Perhaps the larger concern is with food security since so much of our food production depends on irrigated agriculture, especially in Arizona and Southern California – the winter fruit and salad bowls of the nation.

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Further Reading

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